Imaging Features of Premalignant Biliary Lesions and Predisposing Conditions with Pathologic Correlation

Maria Zulfiqar, MD
Deyali Chatterjee, MD
Norihide Yoneda, MD
Mark J. Hoegger, MD
Maxime Ronot, MD, PhD
Elizabeth M. Hecht, MD
Nina Bastati, MD
Ahmed Ba-Ssalamah, MD
Mustafa R. Bashir, MD
Kathryn Fowler, MD

Abbreviations: CCA = cholangiocarcinoma, ERCP = endoscopic retrograde cholangiopancreatography, HASTE = half-Fourier acquisition single-shot turbo spin-echo, H-E = hematoxylin-eosin, IPNB = intraductal papillary neoplasm of the bile duct, MRCP = MR cholangiopancreatography, PSC = primary sclerosing cholangitis, VIBE = volumetric interpolated breath-hold examination

RadioGraphics 2022; 42:1320–1337

https://doi.org/10.1148/rg.210194

Content Codes: CT GI MR

From the Mallinckrodt Institute of Radiology, Washington University School of Medicine, 510 S Kingshighway Blvd, St Louis, MO 63110 (M.Z., M.J.H.); Department of Pathology, University of Texas MD Anderson Cancer Center, Houston, Tex (D.C.); Department of Radiology, Kanazawa University Graduate School of Medical Sciences, Kanazawa, Japan (N.Y.); Department of Radiology, Hôpital Beaujon, APHP. Nord, Clichy & Université de Paris, Paris, France (M.R.); Department of Radiology, Weill Cornell Medicine, New York, NY (E.M.H.); Department of Biomedical Imaging and Image-guided Therapy, Medical University of Vienna, General Hospital of Vienna (AKH), Vienna, Austria (N.B., A.B.S.); Departments of Radiology and Medicine, Duke University Medical Center, Durham, NC (M.R.B.); and Department of Radiology, UC San Diego School of Medicine, San Diego, Calif (K.F.). Recipient of a Cum Laude award for an education exhibit at the 2020 RSNA Annual Meeting. Received July 18, 2021; revision requested September 3 and received September 26; accepted October 4. For this journal-based SA-CME activity, the authors A.B.S., M.R.B., and K.F. have provided disclosures (see end of article); all other authors, the editor, and the reviewers have disclosed no relevant relationships. **Address correspondence to** M.Z. (e-mail: *mariazulfiqar@wustl.edu*).

Biliary malignancies include those arising from the intrahepatic and extrahepatic bile ducts as well as the gallbladder and hepatopancreatic ampulla of Vater. The majority of intrahepatic and extrahepatic malignancies are cholangiocarcinomas (CCAs). They arise owing to a complex interplay between the patient-specific genetic background and multiple risk factors and may occur in the liver (intrahepatic CCA), hilum (perihilar CCA), or extrahepatic bile ducts (distal CCA). Biliary-type adenocarcinoma constitutes the most common histologic type of ampullary and gallbladder malignancies. Its prognosis is poor and surgical resection is considered curative, so early detection is key, with multimodality imaging playing a central role in making the diagnosis. There are several risk factors for biliary malignancy as well as predisposing conditions that increase the risk; this review highlights the pertinent imaging features of these entities with histopathologic correlation. The predisposing factors are broken down into three major categories: *(a)* congenital malformations such as choledochal cyst and pancreaticobiliary maljunction; *(b)* infectious or inflammatory conditions such as parasitic infections, hepatolithiasis, primary sclerosing cholangitis, and porcelain gallbladder; and *(c)* preinvasive epithelial neoplasms such as biliary intraepithelial neoplasm, intraductal papillary neoplasm of the bile duct, intra-ampullary papillary tubular neoplasm, and intracholecystic papillary neoplasm of the gallbladder. Recognizing the baseline features of these premalignant biliary entities and changes in their appearance over time that indicate the advent of malignancy in high-risk patients can lead to early diagnosis and potentially curative management.

An invited commentary by Volpacchio is available online. Online supplemental material is available for this article.

SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Describe common premalignant biliary entities.
- Identify imaging features indicative of malignant transformation.
- Understand premalignant cystic neoplasms in the liver.

See rsna.org/learning-center-rg.

Introduction

The biliary tract is divided into intrahepatic and extrahepatic segments, with their blood supply exclusively from branches of the hepatic artery. The right and left intrahepatic ducts join at the extrahepatic hilum to form the common hepatic duct. The cystic duct drains into the common hepatic duct to form the common bile duct.

Histologically, biliary epithelial cells or cholangiocytes line the biliary tract (1). The biliary epithelial cells lining the smaller intrahepatic ducts (canals of Hering ductules, interlobular and septal)

TEACHING POINTS

- Chronic biliary inflammation predisposes to biliary malignancy.
- Progressive biliary duct dilatation in a patient with Caroli disease should prompt a careful search for an underlying stricture or mass.
- This functional abnormality can lead to recurrent inflammation of the biliary epithelium and increased cellular proliferation, predisposing to biliary tract and gallbladder cancers that develop 15–20 years earlier than in subjects without PBM.
- A progressively worsening stricture, disproportionate focal biliary dilatation, asymmetric biliary thickening, nodularity, or an enhancing filling defect should raise suspicion for malignancy in RPC.
- MCN-L typically does not communicate with the bile ducts, unlike IPNB, which may communicate.

demonstrate proliferative and reparative functions that are normally quiescent but are activated under stress, such as infection, acute or chronic inflammation, or vascular or direct injury (2). Within the larger bile ducts, the biliary epithelial cells participate in hormone-regulated bile excretion and mucin secretion.

Peribiliary glands are tubule-alveolar glands with acini of serous and mucinous cholangioctyes that are located along the wall of bile ducts and exist in various maturational stages, harboring hepatic stem/progenitor cells (HPCs) that can differentiate into mature lineages of cholangiocellular, hepatocellular, or pancreatic endocrine cells. The highest density of peribiliary glands is found at the branching points of the intrahepatic and extrahepatic biliary tract (including the ampulla of Vater and cystic duct origin) (3).

Carcinogenesis

Biliary malignancy can arise at different sites along the biliary tract; depending on the location and type of tumor, its morphologic appearance may vary. There is emerging evidence that in addition to arising from the mature luminal epithelial lining, malignancy can also originate from the peribiliary glands, including the HPCs (4).

Chronic biliary inflammation predisposes to biliary malignancy. Ongoing or recurrent infection or inflammation leads to cholestasis, which irritates the biliary epithelium and peribiliary glands. Cellular injury activates repair mechanisms with cellular proliferation, which eventually leads to dysregulation of DNA repair and apoptosis, leading to survival of mutated cells with onset of metaplasia and dysplasia, biliary intraepithelial neoplasia, and eventually cholangiocarcinoma (CCA) (5).

Although CCA can arise at any point along the biliary tree, the more common sites of biliary malignancy are located where peribiliary glands are highest in density: branching points such as the cystic duct, hepatic hilum, and periampullary region (4). The prevalence of CCA is increasing globally, accounting for nearly 15% of all primary liver cancers and approximately 3% of gastrointestinal malignancies with 2% of all cancer-related deaths every year (6).

Premalignant Biliary Conditions

Several cholangiopathies and biliary lesions increase the risk of malignancy and can be divided into three basic categories, as summarized in the Table.

Congenital Anomalies

Ductal Plate Malformations

Ductal plate malformations are a heterogeneous group of disorders involving a variable degree of malformation and dilatation of the intrahepatic bile ducts and can be associated with hepatic fibrosis from persistent and aberrant remodeling of the embryonal ductal plate. These disorders include choledochal cysts and Caroli disease, autosomal recessive polycystic kidney disease, autosomal dominant polycystic kidney disease, congenital hepatic fibrosis, and rarely von Meyenburg complexes (7). Chronic inflammation with continuous injury of the bile ducts over decades along with recurrent episodes of cholangitis and biliary stones lead to malignancy in 6%–14% of cases, with the overall risk of CCA increased by 100-fold compared with that of healthy individuals (8).

Choledochal Cysts

Choledochal cysts are pathologic dilatations of the biliary tract and are classified into five types on the basis of their anatomic location (Todani classification) (Fig E1). There is increased risk of CCA that escalates with age, from 2.3% for the 3rd decade of life to 75% for the 7th decade (9). A dilated patulous bile duct leads to stasis and recurrent cholangitis, predisposing individuals to carcinogenesis. Adenocarcinoma is most common, accounting for 73%–84% cases, followed by anaplastic carcinoma (10%), undifferentiated carcinoma (5%–7%), squamous cell carcinoma (5%), and other rare tumor types such as neuroendocrine carcinoma (1.5%) (10). The incidence of cancer per Todani type of choledochal cyst is highest with type I choledochal cysts (68%), followed by type IV (21%), type V (6%), type II (5%), and type III (1.6%) (11).

Although surgical resection decreases the risk of malignancy, there remains an increased risk compared with that of the general population in all forms of choledochal cysts. Postexcision malignant disease has been reported in 0.7%–6% of patients. Ten Hove et al (12) showed a four times higher risk of malignancy in patients treated with partial cyst excision than in those treated with complete excision. Malignancy can develop in the residual cyst, at the anastomosis, or in the pancreas. Periodic imaging follow-up for these patients is therefore necessary. Although no consensus is available regarding the frequency of follow-up, annual measurement of serum liver enzyme levels and imaging is usually performed, with the clinical picture driving the need for more frequent imaging.

MRI including cholangiopancreatography (MRCP) without and with intravenous contrast material is the preferred imaging modality for identification and surveillance of individuals with choledochal cysts before and after surgery. It allows accurate classification of the type of the choledochal cyst and identification of the presence and extent of malignancy. Malignancy often manifests as focal or diffuse irregular duct wall thickening with or without a discrete mass. The affected bile duct often demonstrates avid enhancement in the region of thickening on contrast-enhanced images with a similar appearance at CT (13).

A potential pitfall is focal wall thickening secondary to cholangitis, which can be mistaken for a neoplasm (14). As both inflammation and malignancy can show enhancement and diffusion restriction, with absolute differentiation often not possible at imaging, tissue sampling is needed. Morphologically, the tumor may manifest as a T2-intermediate to -hypointense eccentric filling defect in the bile duct lumen or in a periductal pattern surrounding the duct wall (Fig 1).

Caroli Disease

Caroli disease is an autosomal recessive disorder arising owing to ductal plate malformation, with cystic dilatation of the intrahepatic bile ducts that can be segmental (83%) or diffuse (17%). The degree of dilatation is variable and can be linear, fusiform, or saccular. One important feature is the absence of liver dysmorphia or segmental atrophy in Caroli disease (15). Presence of the central dot sign is pathognomonic, representing the portal vein surrounded by dilated bile ducts (16).

A potential pitfall is the presence of well-demarcated cysts along the bile ducts with varying sizes in a hilar distribution on both sides of the portal vein, which represent peribiliary cysts. These are in fact benign retention cysts of the peribiliary glands that arise in the setting of severe chronic liver disease and portal hypertension and should not be confused with cystic biliary dilatation or neoplasm. These are often incidental and require no radiologic follow-up (17) (Fig E2).

At imaging, CCA can take many forms in patients with Caroli disease. Worsening focal bile duct dilatation and capsular retraction can be seen with or without an irregular intraparenchymal mass (Fig 2). When present, the mass often shows peripheral continuous enhancement in the arterial phase with delayed progressive central enhancement (18). Progressive bile duct dilatation in a patient with Caroli disease should prompt a careful search for an underlying stricture or mass.

Predisposing Conditions and Biliary Lesions with Increased Risk for Biliary Malignancy

Congenital anomalies
Choledochal cysts and Caroli disease
Pancreaticobiliary maljunction (PBM)
Autosomal dominant and recessive polycystic kidney disease
Congenital hepatic fibrosis
Von Meyenburg complexes
Inflammatory or infectious conditions
Primary sclerosing cholangitis (PSC)
Recurrent pyogenic cholangitis
Hepatolithiasis
Chronic viral hepatitis (C and B)
Cystic fibrosis cholangitis
Porcelain gallbladder
Chronic cholecystitis
Cholelithiasis
Neoplasms
Biliary intraepithelial neoplasm (BilIN)
Adenofibroma
Intraductal papillary neoplasm of the bile duct (IPNB)
Mucinous cystic neoplasm (MCN)
Intra-ampullary papillary tubular neoplasm (IAPN)
Intracholecystic papillary neoplasm (ICPN)

Pancreaticobiliary Maljunction

Pancreaticobiliary maljunction (PBM) is a congenital anomaly where the junction of the pancreatic and bile ducts is located outside the duodenal wall, forming a common channel with length greater than 1 cm (Fig 3). This causes decreased effectiveness of the sphincter of Oddi in regulation of outflow of bile and pancreatic secretions, leading to stasis and reflux of pancreatic juices into the biliary tract. This functional abnormality can lead to recurrent inflammation of the biliary epithelium and increased cellular proliferation, predisposing to biliary tract and gallbladder cancers that develop 15–20 years earlier than in subjects without PBM (19).

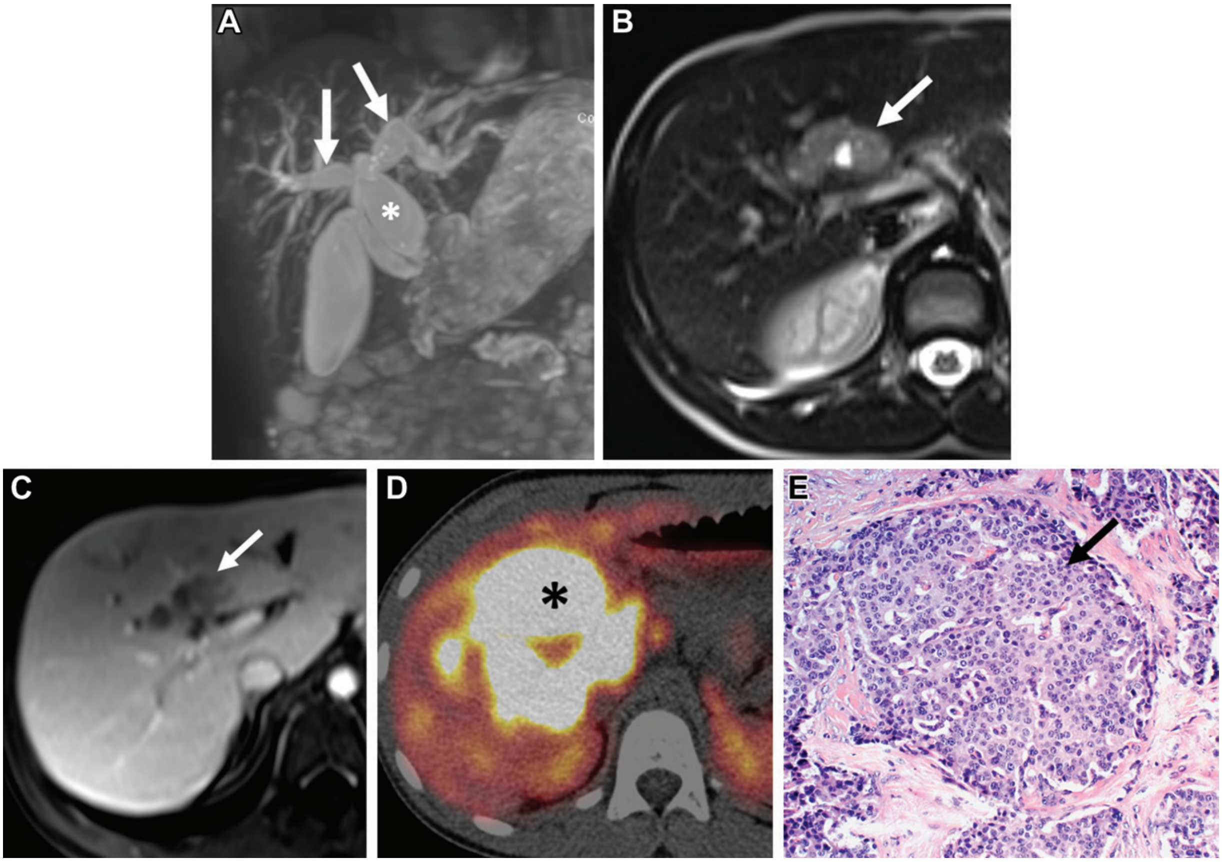

Figure 1. Neuroendocrine biliary neoplasm arising 2 years after partial excision of a type IVa choledochal cyst in an 8-year-old girl. **(A)** Coronal MRCP image shows cystic dilatation of the common bile duct (*) that also involves the right and left hepatic ducts (arrows). Excision of the extrahepatic component with a hepaticojejunostomy was performed. There was microscopic neuroendocrine tumor (NET) in the histopathologic specimen. **(B)** Follow-up axial T2-weighted half-Fourier acquisition single-shot turbo spin-echo (HASTE) image shows a T2-isointense periductal mass (arrow) at the anastomosis. **(C)** Axial contrast-enhanced portal venous T1-weighted volumetric interpolated breath-hold examination (VIBE) image shows enhancement within the mass (arrow). **(D)** Somatostatin receptor gallium 68–DOTATATE PET/CT image 8 months later shows progression of the NET (*), with increased size and invasion into the hepatic parenchyma. **(E)** Photomicrograph of the tumor shows clusters of neuroendocrine cells with round to oval nuclei (arrow). (Hematoxylin-eosin [H-E] stain; original magnification, ×40.)

The risk of malignancy is variable, depending on whether concomitant biliary dilatation is present (77%) or not (23%). Choledochal cysts often coexist with PBM and frequently involve the common bile duct rather than the intrahepatic ducts (20). When PBM is associated with biliary dilatation, the risk for bile duct cancer is 6.9% and that for gallbladder cancer is 13.4%. PBM without biliary dilatation increases the risk of bile duct cancer to a lesser degree (3.1%), but the risk of gallbladder cancer is increased to 37.4%, suggestive of a protective effect of biliary dilatation (19).

MRCP is the imaging modality of choice for identifying the presence of PBM and evaluating duct diameter and length of PBM, associated choledochal cysts, and development of malignancy. One pitfall to keep in mind is that narrowing of the distal duct near the PBM or hilum is a common finding in these patients and should not be confused with malignant stricture. This strictured appearance is often relative to the dilated parts of the choledochal cyst.

However, increasing duct narrowing with associated wall thickening and enhancement should raise concern for malignancy, and tissue sampling may be warranted (21). Owing to chronic inflammation, gallbladder wall thickness is often increased in patients with PBM (6 mm vs 3 mm in normal cohorts). Asymmetric gallbladder wall thickening or nodular or masslike soft tissue at imaging should raise concern for malignancy (Fig 4). Prophylactic cholecystectomy is often performed in patients with isolated PBM, with suggested yearly follow-up with measurement of liver enzyme levels and MRCP (22).

Inflammatory or Infectious Conditions

Primary Sclerosing Cholangitis

Primary sclerosing cholangitis (PSC) is an idiopathic chronic inflammatory disease of the bile ducts with progressive inflammatory destruction of the bile ducts. The exact mechanism of PSC is not clearly understood but is favored to

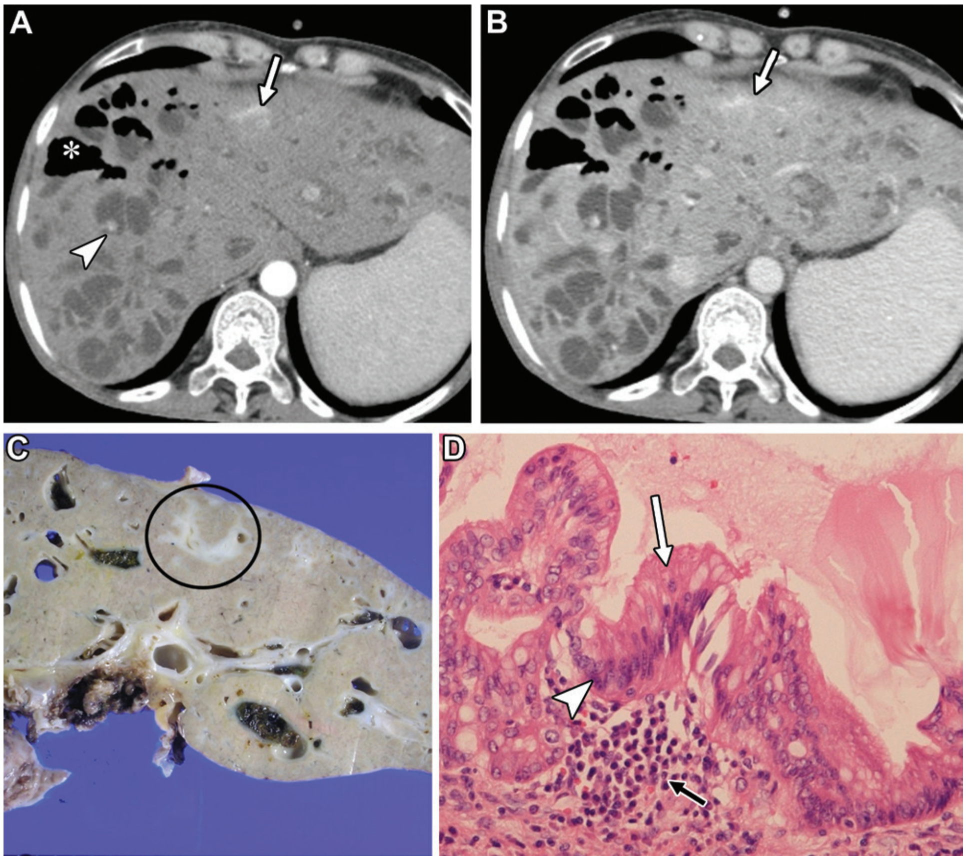

Figure 2. Caroli disease and early intrahepatic CCA in a 58-year-old woman with abdominal pain. **(A)** Axial contrast-enhanced arterial phase CT image shows diffuse cystic dilatation of the bile ducts in the liver, most pronounced on the right (*), which represents Caroli disease. The central portal vein (arrowhead) is observed inside the dilated bile ducts, consistent with the central dot sign. There is gas in the bile ducts owing to recent endoscopic retrograde cholangiopancreatography (ERCP). In the left hemiliver, there is an ill-defined masslike area of arterial enhancement (arrow). **(B)** Axial contrast-enhanced portal venous phase CT image shows more confluent delayed enhancement in this region (arrow), resembling that of focal cholangitis. Liver transplant was performed. **(C)** Gross pathologic specimen of the explant shows a white-toned fibrotic-appearing abnormal mass (circle) in the same area, which corresponds to the findings at CT. **(D)** Photomicrograph shows extensive peribiliary inflammatory cell infiltration (black arrow) with biliary intraepithelial neoplasm (BilIN) grade 3 (carcinoma in situ), as evidenced by the pseudopapillary architecture of the biliary mucosa (white arrow) and severe nuclear irregularities (arrowhead). (H-E stain; original magnification, ×40.)

be autoimmune. Approximately 70% of cases are seen with concomitant inflammatory bowel disease (23).

MRCP is indispensable for initial diagnosis and follow-up in patients with PSC. Given the need for serial follow-up over a long period, noninvasive imaging that does not require radiation is favored. Multifocal areas of short-segment biliary narrowing occur owing to periductal sclerosis and fibrosis with normal-caliber or mildly dilated intervening ducts, giving a characteristic beaded appearance. Linear biliary wall enhancement and wedge-shaped areas of transient hepatic intensity difference (THID) or transient hepatic attenuation difference (THAD) in the hepatic parenchyma on contrast-enhanced MR or CT images correspond to acute inflammation (better seen in the arterial phase) or venous abnormalities and fibrosis (better seen in the portal venous phase) (24).

PSC has significantly increased risk of bile duct and gallbladder malignancy as well as hepatocellular carcinoma (HCC). Compared with CCA in the general population, CCA in PSC involves a younger demographic in the 4th decade of life, as the risk for CCA is 160–1560 times higher (25). The phenotype of CCA seen in PSC is commonly periductal infiltrating, which makes early recognition at cross-sectional imaging somewhat challenging. In the setting of PSC, intrahepatic CCA is more common than extrahepatic CCA (<10%).

The concept of the dominant stricture is derived from endoscopic retrograde cholangiopancreatography (ERCP), with an intraluminal diameter of less than 1.5 mm in the common bile duct and less than 1 mm in the hepatic ducts. As ERCP is performed with pressure injection, the degree of distensibility of the duct can vary compared with that at MRCP. More descriptive

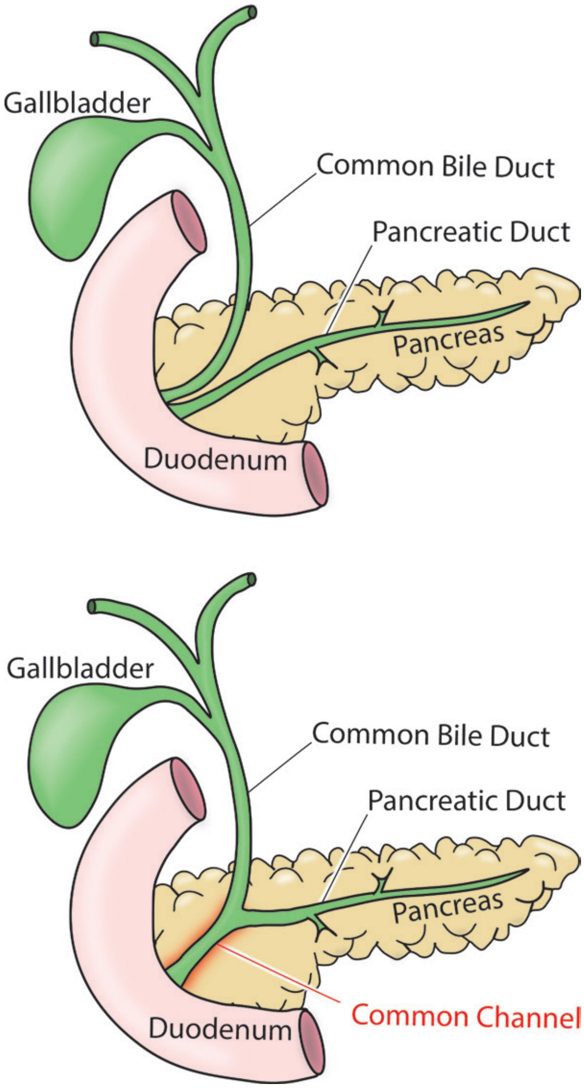

Figure 3. Comparison of normal pancreaticobiliary anatomy (top) and PBM (bottom). With PBM, note the abnormal long length and extramural location of the common channel, which is situated within the pancreatic parenchyma and facilitates reflux of enzymatic pancreatic secretions into the biliary tract and gallbladder.

reporting of the luminal narrowing at MRCP is encouraged as mild, moderate, or severe along with length of involvement and presence or absence of associated biliary wall thickening, enhancement, or ductal dilatation (26). A new or worsening irregular high-grade stricture at MRCP should be investigated with ERCP and brushing (Fig 5).

There may or may not be an associated mass, but focal biliary wall thickening and enhancement may be present. Shouldered margins of the stricture and rapid progression of bile duct narrowing may be other subtle clues toward an underlying malignancy. Segmental or subsegmental moderate to severe bile duct dilatation should prompt careful evaluation for an obstructing mass or intraluminal polypoid lesions, which may be subcentimeter in the early stages. Mass-forming intrahepatic CCA manifests as a hepatic mass with or without capsular retraction and progressive delayed enhancement with varying degrees of hepatic parenchymal infiltration (27).

Recurrent Pyogenic Cholangiohepatitis

Recurrent pyogenic cholangiohepatitis (RPC) is a chronic parasitic infection of the bile ducts in endemic areas such as Southeast Asia owing to infestation by *Clonorchis sinensis* or *Ascaris lumbricoides*, which causes biliary injury with recurrent bouts of bacterial cholangitis, leading to multifocal strictures (28). Direct biliary injury or stricture formation promotes hepatolithiasis and superinfection with enteric organisms, particularly *Escherichia coli*, which are cultured from bile and blood in the majority of these patients. The bacteria deconjugates bilirubin glucuronide, which then precipitates with calcium in the bile as brown friable calcium bilirubinate stones (29).

Intrahepatic stones can be delineated well at US and CT as echogenic foci with posterior acoustic shadowing and hyperattenuating filling defects, respectively, in the dilated irregular bile ducts with multifocal biliary strictures. The need for imaging in RPC is primarily driven by the patient's symptoms and liver enzyme levels. MRCP is superior to CT or US for visualization of intraductal stones. There is dilatation of the first- and second-order bile ducts with intraluminal stones that are T2 hypointense and T1 hyperintense owing to bilirubin content. These stones are recurrent and often nonobstructive with nondilated upstream peripheral ducts.

Biliary strictures are often short segment and manifest at MRCP as decreased arborization and abrupt tapering of the peripheral bile ducts, termed the *arrowhead sign* (Fig E3). Chronic periductal inflammation leads to segmental or diffuse parenchymal injury and fibrosis (30). Multidisciplinary management is warranted and depends on the severity of disease, including antibiotic therapy for acute attacks, endoscopic dilation of strictures and stone removal, biliary drainage, partial hepatectomy, or in severe cases liver transplantation (28).

The risk for CCA is increased by 3%–10% in patients with RPC. Malignancy tends to arise in atrophied segments or ducts with heavy stone burden, manifesting as focal ductal narrowing and thickening. A progressively worsening stricture, disproportionate focal biliary dilatation, asymmetric biliary thickening, nodularity, or an enhancing filling defect should raise suspicion for malignancy in RPC. Mass-forming intrahepatic CCA associated with RPC can be necrotic, making it difficult to differentiate from a hepatic

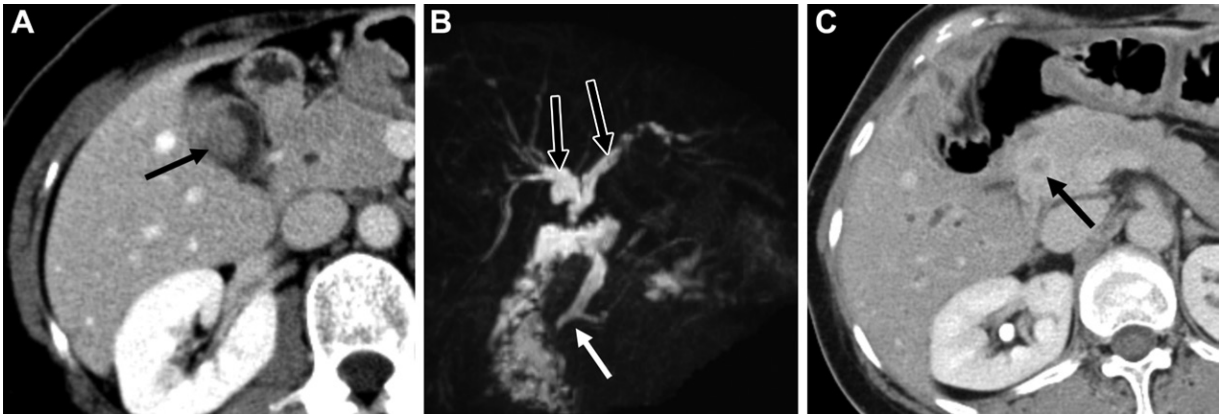

Figure 4. Gallbladder and bile duct malignancy in a 55-year-old woman with anomalous PBM and Todani type IVa choledochal cyst. **(A)** Axial contrast-enhanced portal venous phase CT image shows an enhancing mass (arrow) in the gallbladder lumen, concerning for malignancy, which was confirmed at pathologic analysis after cholecystectomy. **(B)** Coronal MRCP image shows a nondilated common channel (white arrow) and intrahepatic component (black arrows) of the choledochal cyst after hepaticojejunostomy. **(C)** Axial contrast-enhanced portal venous phase CT image 3 years later shows wall thickening and enhancement in the residual intrapancreatic bile duct (arrow), which was confirmed to be CCA at pathologic analysis.

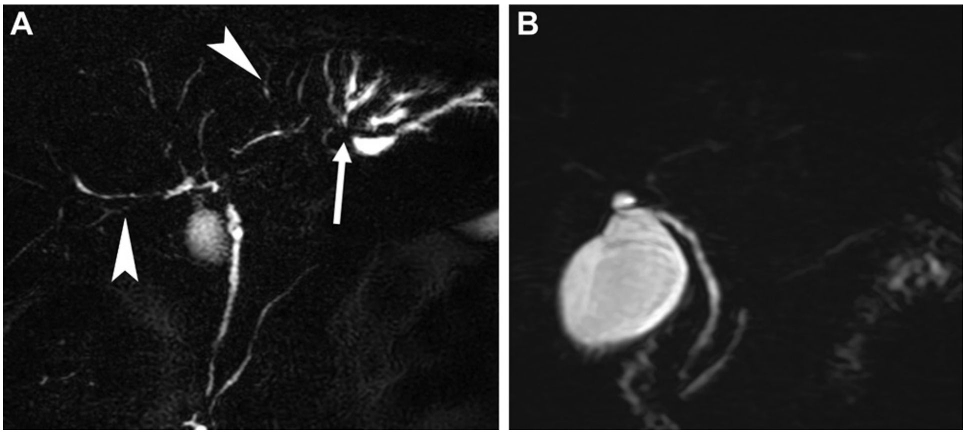

Figure 5. PSC complicated by CCA in a 40-year-old man with ulcerative colitis. **(A)** Coronal MRCP image of the liver shows multifocal intrahepatic biliary strictures with a beaded appearance (arrowheads), representing changes of PSC. There is a high-grade stricture (arrow) in the left hemiliver with moderate upstream intrahepatic bile duct dilatation. **(B)** Coronal MRCP image of the liver 1 year earlier shows mild contour irregularity of the bile ducts but no high-grade stricture or bile duct dilatation. ERCP with brushing of the stricture was performed; pathologic analysis demonstrated CCA.

abscess, requiring tissue sampling for confirmation (31). Less commonly, CCA may manifest with marked mucin secretion, leading to disproportionate biliary dilatation (32) (Fig 6).

Chronic Hepatitis

Chronic hepatitis such as viral or autoimmune hepatitis with or without cirrhosis has been shown to increase risk for CCA. Chronic hepatic injury stimulates the hepatic stem/progenitor cells (HPCs) to proliferate, which eventually leads to carcinogenesis. The postulated origin from HPCs supports why mass-forming CCA is more common in the setting of chronic hepatitis compared with periductal and intraductal growth–type CCA, which primarily originates from biliary epithelial cells (33). Cirrhosis with chronic hepatitis is believed to exert a synergistic effect on CCA development, with CCA occurring nearly 10 years earlier in patients with hepatitis B than in those with hepatitis C (34).

The majority of CCAs are detected as part of surveillance imaging in patients with chronic hepatitis B or with cirrhosis due to hepatitis C or autoimmune hepatitis. In the appropriate patient demographic, use of the Liver Imaging Reporting and Data System (LI-RADS) CT and MR criteria helps distinguish hepatocellular carcinoma (HCC) from CCA, although both can share common features. The diagnosis of combined HCC-CCA is more challenging, a significant subset of the latter displaying features of CCA (35).

At MRI and CT, rim arterial phase hyperenhancement (APHE) with central enhancement

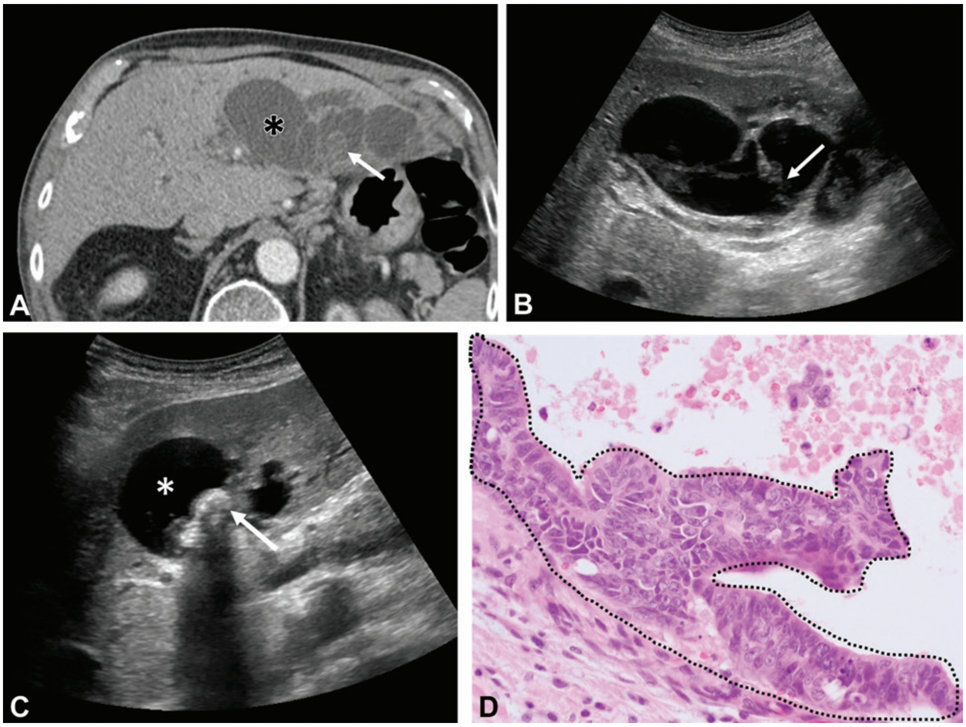

Figure 6. Recurrent pyogenic cholangitis in a 73-year-old man with worsening epigastric pain. **(A)** Axial contrast-enhanced portal venous CT image shows varicose biliary dilatation (*) in the left lateral section of the liver, which had progressively worsened in the past year. There is a mildly attenuating filling defect (arrow) in the dilated duct. **(B)** Transverse US image of the liver shows irregular thick walls (arrow) of the dilated ducts. **(C)** US image of the liver shows intraluminal stones (arrow), which correspond to the filling defect at CT. Owing to marked disproportionate ductal dilatation and wall thickening, suspicion was raised for a possible underlying intraductal papillary neoplasm of the bile duct (IPNB). * = varicose biliary dilatation. **(D)** Photomicrograph obtained after left hepatectomy shows an IPNB with high-grade dysplasia (dotted outline). (H-E stain; original magnification, ×200.)

in subsequent delayed phases in a progressive concentric pattern or a peripheral washout appearance strongly favors CCA over HCC, leading to categorization of the lesion as LR-M in LI-RADS (Fig 7). Additional features of CCA include capsular retraction, biliary obstruction disproportionate to that expected on the basis of mass size, and target appearance at diffusion-weighted imaging (DWI) or in the hepatobiliary phase when using a hepatobiliary contrast agent (36,37). Although less common than with HCC, enhancing tumor thrombus can be seen with CCA extending into the portal vein or bile ducts or both and when present indicates a worse prognosis (38).

Cystic Fibrosis

Cystic fibrosis is an autosomal recessive multiorgan disorder involving the cystic fibrosis transmembrane conductance regulator *(CFTR)* gene, mutations of which lead to generalized exocrine dysfunction. The *CFTR* gene is highly expressed throughout the biliary system, including the gallbladder. As the gene is lost, there is decreased biliary bicarbonate secretion, which promotes reduced bile flow and increased sensitivity to inflammatory toxins, with subsequent periductal inflammation and proliferation promoting periportal fibrosis. Progressive liver disease occurs in about 10% of patients with cystic fibrosis and represents the third most common cause of death in these patients (39).

The morphologic findings at imaging are diverse and include changes in the liver such as steatosis to a PSC-like appearance with multifocal intra- and extrahepatic biliary strictures. Up to 30% of cystic fibrosis patients develop advanced liver disease, including biliary cirrhosis and portal hypertension, which aggravate the respiratory dysfunction and increase risk for early mortality. Focal biliary cirrhosis has been reported in 20%–30% of cystic fibrosis patients and is considered the pathognomonic hepatic lesion, characterized by thickened hyperechoic periportal tissue greater than 2 mm at US and high signal intensity at T1-weighted MRI, corresponding histologically to fat globules in the fibrotic portal tracts and large fat-laden vacuoles in the periportal liver parenchymal cells (Fig E4) (40). It can lead to multilobular cirrhosis in 5%–10% of cystic fibrosis patients, where the liver parenchyma is replaced by regenerative nodules (41).

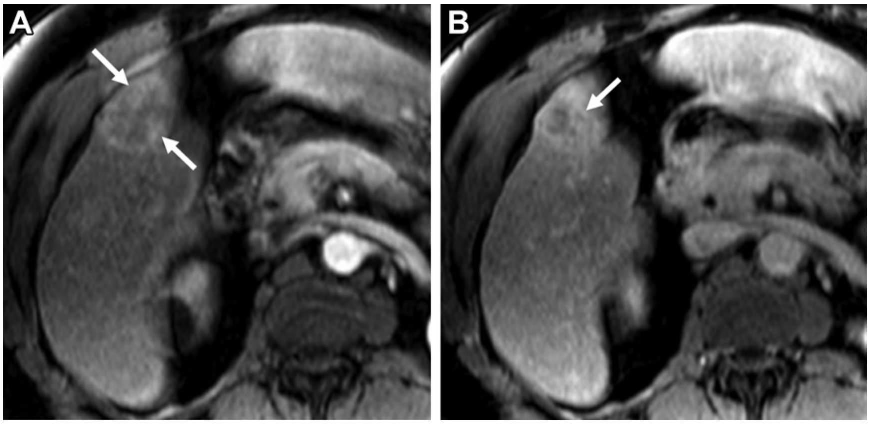

Figure 7. Intrahepatic CCA in a 63-year-old man with hepatitis C cirrhosis. **(A)** Axial contrast-enhanced arterial phase T1-weighted VIBE image shows rim arterial phase hyperenhancement (APHE) in a mass (arrows) in hepatic segment V, meeting LI-RADS criteria for LR-M. **(B)** Axial contrast-enhanced delayed phase T1-weighted VIBE image shows progressive central enhancement in the mass (arrow). Biopsy was performed; histopathologic analysis demonstrated CCA.

Inspissated biliary secretions predispose to development of microgallbladder in 4%–45% of cases, possibly through stenosis of the cystic duct (42). Chronic biliary inflammation in cystic fibrosis leads to stones and sclerosing cholangitis and predisposes to CCA (Fig 8). Patients with cystic fibrosis have a 17-fold increased risk of biliary malignancy in comparison with the general population; the risk increases two- to fivefold after organ transplant. Serial MRI should be considered in these patients with large bile duct strictures and worsening cholestasis (43).

Porcelain Gallbladder

The term *porcelain gallbladder* or *cholecystopathia chronica calcarea* refers to the blue discoloration and brittle consistency of the gallbladder wall at surgery due to dense fibrosis, which invariably occurs owing to chronic gallbladder inflammation and associated gallstones in 95% of cases. At imaging, there is extensive calcific encrustation of the gallbladder wall, which often appears as a band of calcium conforming to the shape of the gallbladder (Fig 9). Albeit less than previously believed, there is still increased risk of gallbladder cancer in the context of porcelain gallbladder, which has been reported as 1%–6% (44). Prophylactic cholecystectomy is the treatment of choice; however, periodic imaging follow-up is a reasonable option for asymptomatic patients or those at high surgical risk (45).

Patients with focal mucosal calcification may be at higher risk for cancer than those with a complete pattern of calcification (46). Focal or diffuse gallbladder wall thickening or presence of a mass in the setting of porcelain gallbladder is concerning for malignancy and should prompt the radiologist to scrutinize the adjacent hepatic parenchyma for local invasion. The surrounding peritoneum should also be carefully evaluated for peritoneal carcinomatosis.

Neoplasms

Two main classes of biliary precursor lesions are proposed for development of CCA: biliary intraepithelial neoplasia and papillary neoplasms (Fig 10). Adenofibroma and mucinous cystic neoplasm (MCN) are additional premalignant biliary entities (47).

Bile Ducts

Biliary Intraepithelial Neoplasm.—Previously called biliary atypia or dysplasia, biliary intraepithelial neoplasm (BilIN) is a flat micropapillary lesion, histologically composed of a spectrum of atypical epithelial cells ranging from low-grade lesions to carcinoma in situ (48). Often present adjacent to invasive CCA, BilIN is believed to be the precursor lesion in the dysplasia-to-carcinoma sequence and is strongly associated with chronic inflammatory conditions, such as hepatolithiasis, recurrent pyogenic cholangiohepatitis (RPC), chronic viral hepatitis, and PSC. BilIN has been reported to be present in nearly 58% of cases of CCA, with a prevalence of 75% in extrahepatic CCA compared with 21% in intrahepatic CCA (49).

At imaging, BilIN generally does not have any specific features owing to its microscopic structure and often is not detected prospectively. In our experience, transient peribiliary enhancement can be seen at dynamic CT and MRI, often due to superimposed cholangitis and commonly associated with hepatolithiasis (Fig 11). BilIN is therefore difficult to differentiate from nonneoplastic inflammatory conditions and is almost always an incidental histologic diagnosis

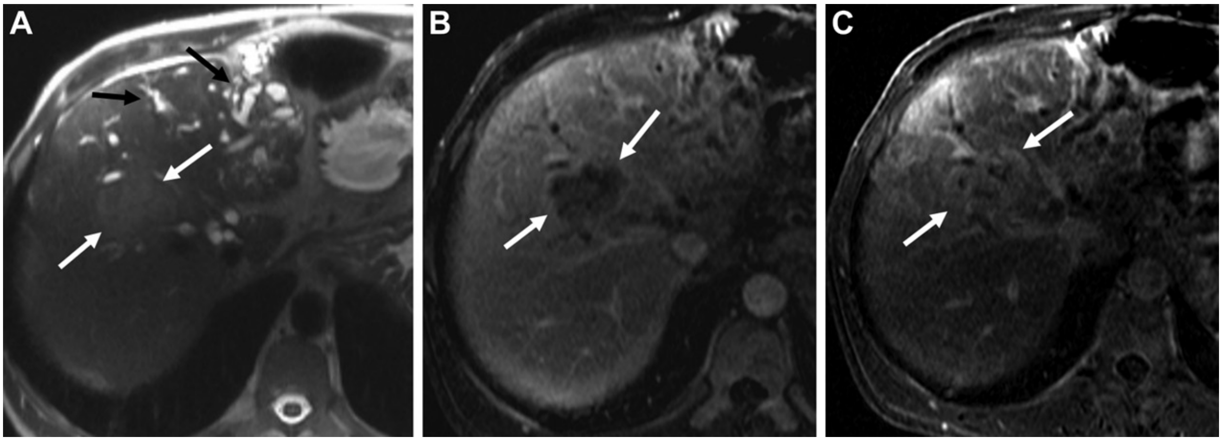

Figure 8. Intrahepatic CCA in a 40-year-old man with cystic fibrosis who underwent lung transplant and presented with increasing bilirubin level. **(A)** Axial T2-weighted HASTE image shows irregular intrahepatic bile duct dilatation with multifocal stricturing (black arrows), most pronounced in the left hemiliver. There is a T2-isointense mass in segment VIII (white arrows). **(B, C)** Axial contrast-enhanced T1-weighted VIBE subtraction images in the arterial phase **(B)** and portal venous phase **(C)** show progressive internal enhancement in the mass (arrows), concerning for CCA, which was confirmed at subsequent biopsy.

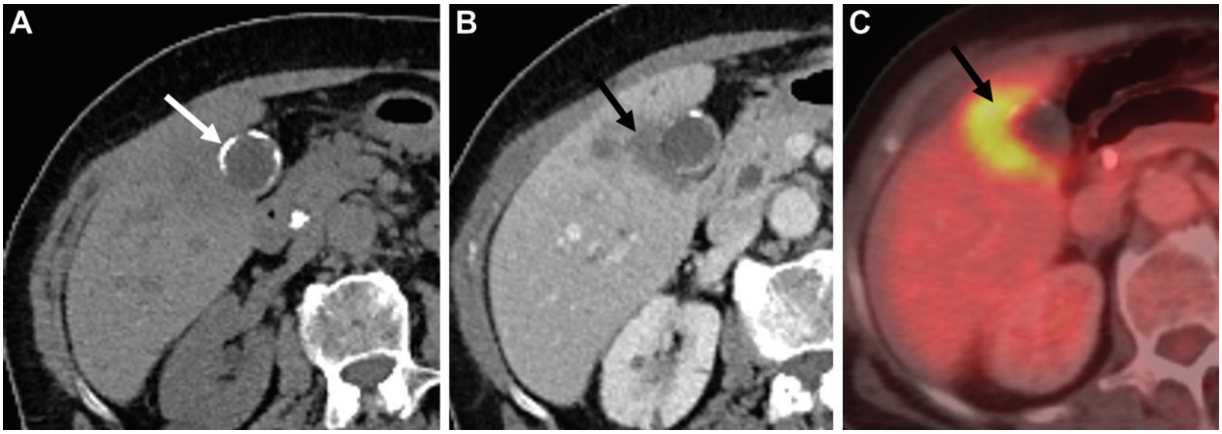

Figure 9. Porcelain gallbladder in a 77-year-old woman. **(A)** Axial image from noncontrast CT performed for vague abdominal pain shows circumferential calcification (arrow) of the gallbladder wall. Two years later, the patient presented with right upper quadrant discomfort. **(B)** Axial contrast-enhanced portal venous phase CT image shows an ill-defined hypoenhancing mass (arrow) arising from the gallbladder, with discontinuity of the wall calcifications and extension of the mass into the adjacent hepatic parenchyma. **(C)** Fluorine 18–fluorodeoxyglucose (FDG) PET/CT image shows that the mass (arrow) is FDG avid, representing adenocarcinoma, which was surgically resected.

in specimens that are removed for other indications, such as CCA, recurrent cholangitis, or cholecystitis (47).

Intraductal Papillary Neoplasm of the Bile Duct.—Intraductal papillary neoplasm of the bile duct (IPNB) is characterized by intraluminal dysplastic epithelial growth that can occur in the intra- or extrahepatic bile ducts. Histologically, papillary frondlike infoldings are seen, composed of columnar epithelial cells around a fibrovascular stalk that connects to the lamina propria (50). IPNB can be low, intermediate, or high grade or associated with invasive carcinoma, depending on the degree of dysplasia at histologic analysis. Excessive mucin production is associated with 30%–40% of IPNBs owing to intestinal metaplasia (51). The vast majority of clinically detectable IPNBs are malignant, but if caught early, these carry a favorable prognosis compared with the mass-forming and periductal infiltrating types of CCA (51).

The imaging features of IPNBs are based on the interplay between formation of intraductal tumors and degree of mucin production as well as tumor location. When tumor formation predominates, an intraductal papillary mass is identified at imaging. Predominant mucin production manifests as ductal dilatation that can be focal, segmental, or diffuse; an associated intraductal tumor may or may not be identified (52). Four different tumor growth patterns can be seen at imaging.

1. The *polypoid intraductal growth pattern* is the classic pattern of IPNB and leads to a polypoid or cauliflower-like mass with micro- or macroscopic papillary projections, akin to the surface

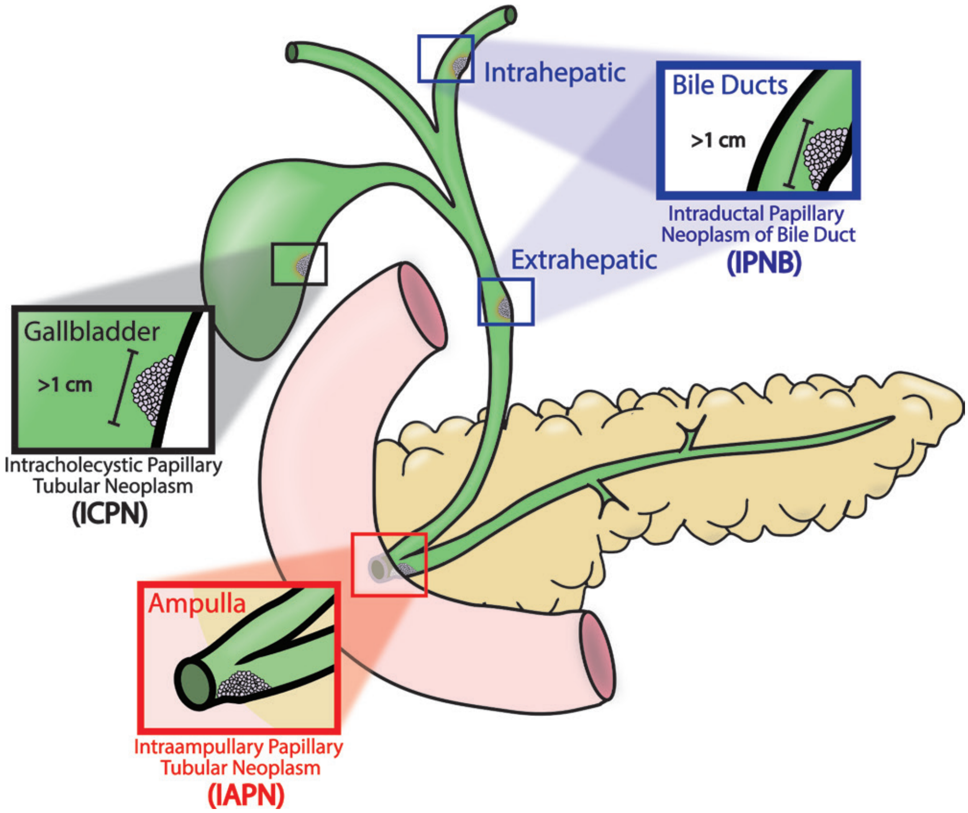

Figure 10. Diagram to elucidate precursor papillary lesions in the bile ducts, ampulla, and gallbladder per the World Health Organization (WHO) 2019 classification of benign and precursor biliary tumors, which describes macroscopically recognizable (>1 cm) papillary neoplastic lesions in the biliary system as counterparts of pancreatic intraductal papillary mucinous neoplasm (IPMN) on a histologic basis. In the bile ducts, the lesion is called intraductal papillary neoplasm of the bile duct (IPNB). In the ampulla, the lesion is called intra-ampullary papillary tubular neoplasm (IAPN). In the gallbladder, the lesion is called intracholecystic papillary tubular neoplasm (ICPN).

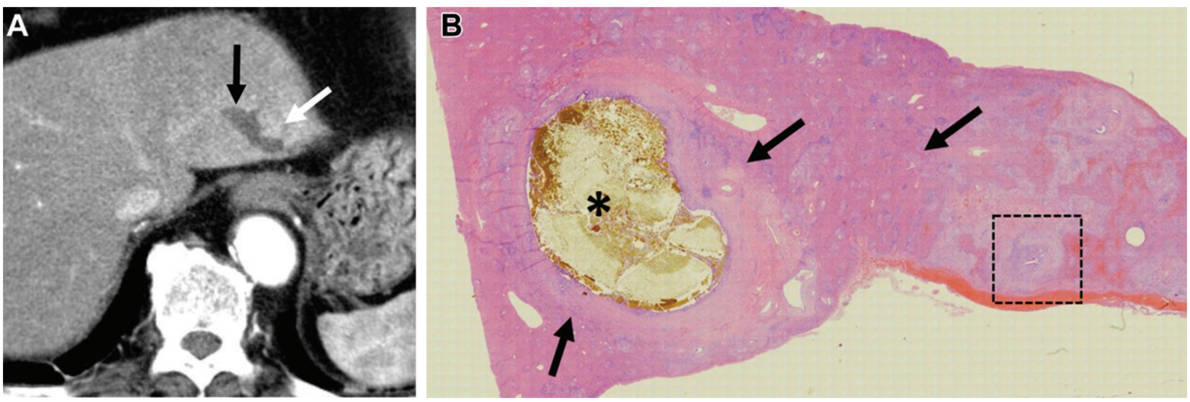

Figure 11. BilIN in a 77-year-old man with abdominal pain. **(A)** Axial contrast-enhanced arterial phase CT image shows focal biliary dilatation (black arrow) in hepatic segment II and mild peribiliary enhancement (white arrow). No definite mass is seen. **(B)** Photomicrograph after left lateral segmentectomy of the liver shows impacted hepatolithiasis (*) and suppurative cholangitis. There is also peribiliary cellular infiltration (arrows), corresponding to the area of enhancement at CT and a focus of BilIN (dashed square) (Fig E5). (H-E stain; original magnification, ×40.)

of a nipple. Upstream bile duct dilatation ensues owing to mechanical obstruction by the mass (Fig 12). T2-weighted MRI is excellent for showing the papillary pattern of the tumor owing to high contrast between the luminal bile and the ingrowing tumor. At contrast-enhanced CT or MRI, the mass demonstrates enhancement greater than or equivalent to that of the hepatic parenchyma in the arterial phase. Scarce fibrous stroma and the presence of a fibrovascular stalk rarely, if ever, allow IPNB to show delayed-phase enhancement, as seen with CCA. Diffusion-weighted imaging

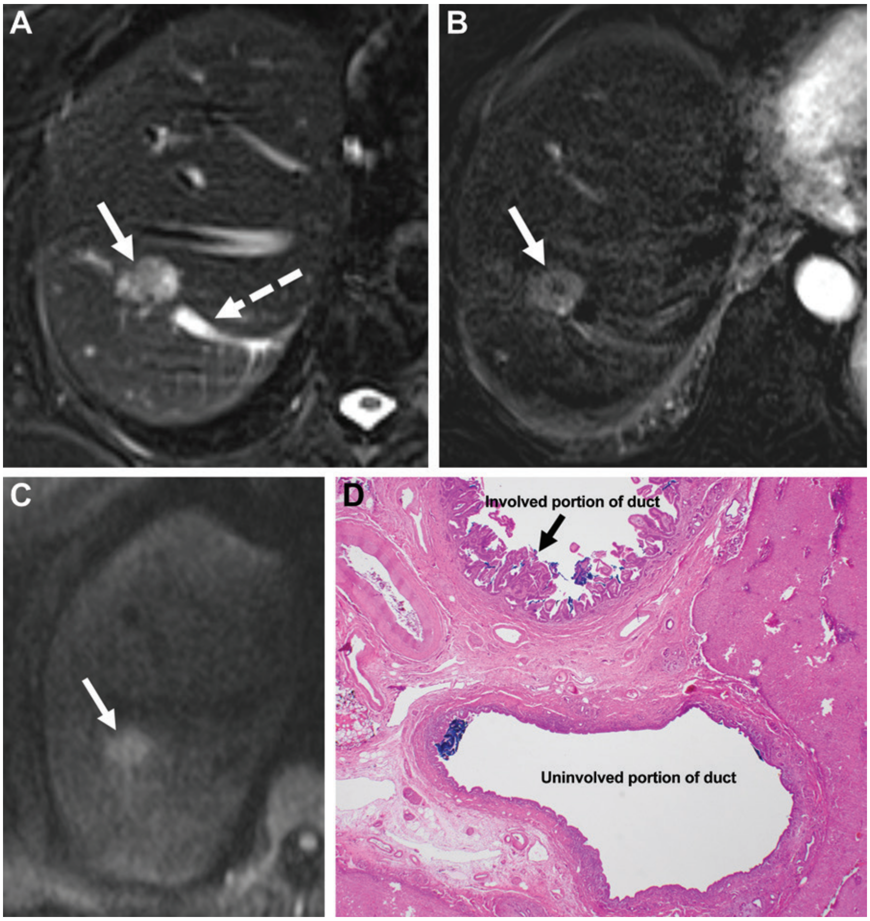

Figure 12. IPNB (polypoid intraductal growth pattern) in an 80-year-old woman with right upper quadrant pain. **(A)** Axial fat-suppressed T2-weighted HASTE image shows a T2-iso-intense filling defect (solid arrow) in the hepatic segment VII bile duct and moderate upstream intrahepatic segmental bile duct dilatation (dashed arrow). **(B)** Axial contrast-enhanced arterial phase subtraction T1-weighted image shows avid enhancement in the mass (arrow). **(C)** Axial diffusion-weighted image shows high signal intensity in the mass (arrow) with corresponding low signal intensity on an apparent diffusion coefficient (ADC) map (not shown). Right hepatectomy was performed. **(D)** Photomicrograph shows papillary projections (arrow) in the lumen of the involved bile duct, which correspond to the mass seen at imaging and represent an IPNB with high-grade dysplasia. (H-E stain; original magnification, ×40.)

(DWI) with the corresponding apparent diffusion coefficient (ADC) improves lesion conspicuity and detection. Decreased ADC near or below 1000×10^{-6} mm^2/sec may be associated with invasiveness (53).

2. The *mucosal spreading growth pattern* lacks the presence of a sizable mass, with the IPNB exhibiting a flat or sessile irregular appearance parallel to the duct wall. In this pattern, the lumen of the bile duct may be carpeted with tiny papillary projections resembling coral reef projections with intervening normal mucosa, also known as biliary papillomatosis. When this growth pattern is associated with massive mucin production, patients present with dilatation of the bile ducts, which is often lobar or segmental with difficult-to-detect IPNB tumor at imaging. Long-standing cases may show parenchymal atrophy of the affected hepatic lobe or segment (47).

3. The *castlike intraductal growth pattern* leads to IPNB tumor growth in the lumen over a considerable length and is of varying thickness. At US, castlike intraductal masses are seen with a preserved uniformly echogenic duct wall. At CT and MRI, the tumor enhances as an intraluminal mass with irregular papillary projections (Fig E6). The presence of the tumor leads to the bile duct lumen appearing rugged and narrowed, but the duct wall is continuous and the periductal soft tissues remain intact unless invasive cancer is also present (54).

4. The *cystic growth pattern* occurs secondary to excessive mucin production by the IPNB, with disproportionate focal aneurysmal dilatation of the involved bile duct from hindered biliary drainage or when the IPNB arises within a peribiliary gland. At imaging, there is a unilocular or multilocular cystic lesion with mural nodularity or

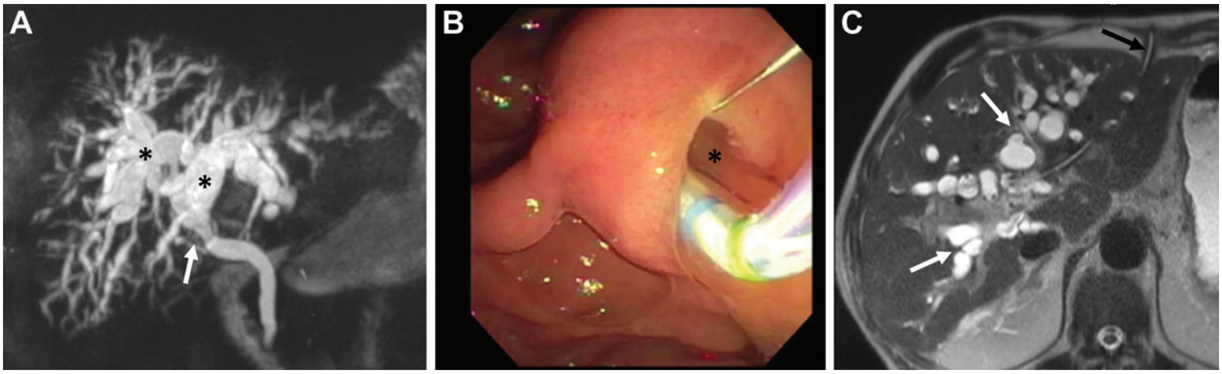

Figure 13. IPNB (mucosal spreading growth pattern) in a 67-year-old man with jaundice. **(A)** Coronal maximum intensity projection (MIP) MRCP image shows asymmetric, T2-hypointense, flat-appearing thickening (arrow) of the bile duct wall at the hilum. There is varicose biliary dilatation (*), the degree of which is disproportionate to the degree of narrowing at the level of the hilar lesion. **(B)** ERCP image shows copious mucin emanating from the ampulla (*). **(C)** Axial T2-weighted HASTE image 6 months later shows worsening biliary dilatation (white arrows) due to viscid mucin even after external biliary stent placement (black arrow). The hilar lesion brushings showed invasive CCA on a background of high-grade dysplasia in the IPNB.

multiple fungating masses along the wall. The cystic tumor may or may not communicate with the bile ducts at cholangiography or hepatobiliary contrast agent MRI (55). A disrupted wall in the region of the IPNB or infiltrated appearance of periductal soft tissues indicates progression to invasive CCA.

The mainstay treatment of IPNB is surgical resection, which requires accurate preoperative assessment, which can often be challenging given the various morphologic growth patterns and mucin production. One pitfall for the radiologist is to mistakenly diagnose obstructed biliary dilatation that in fact is a mucin-distended bile duct secondary to a poorly visible IPNB with a superficial mucosal spreading growth pattern, leading to inadequate drainage with stents or possibly positive resection margins at surgery (Fig 13). Scrutinizing the smoothness of the biliary wall may help with undulations or irregularity of the wall, indicating the presence of an IPNB.

Bile Duct Adenofibroma.—Bile duct adenofibroma is a rare benign bile duct tumor with malignant potential, which is composed of abundant fibrous stroma lined with tubulocystic non–mucin-producing biliary epithelium (56). In the literature, only 22 cases have been reported, eight of which underwent malignant transformation into CCA (57). At CT and MRI, adenofibroma can appear as a multiseptated cystic mass with a thick wall and internal septal enhancement after contrast material administration or—as in our experience—as a well-circumscribed solid enhancing mass with areas of diffusion restriction (Fig 14) (58).

There may be internal areas of necrosis and usually no associated biliary dilatation. There is no communication with the biliary tree. Increasing size and heterogeneity of the mass are concerning for malignant transformation (59). Surgical resection is the treatment of choice.

Mucinous Cystic Neoplasm of the Liver.—Previously referred to as cystadenoma, mucinous cystic neoplasm of the liver (MCN-L) arises from peribiliary glands of the intrahepatic bile ducts, forming a multiloculated cystic lesion in the liver, and typically occurs in women (60). A single layer of cuboidal or tall columnar mucin-producing epithelial cells is present at histologic analysis, below which a band of spindle cells is seen resembling ovarian stroma, the presence of which is a prerequisite for pathologic diagnosis of MCN-L.

The tumor can be low, intermediate, or high grade, depending on the degree of MCN-L epithelial dysplasia, and can have associated invasive carcinoma, which is mostly tubular-type adenocarcinoma (61). At imaging, a benign MCN-L appears as a solitary multiloculated cystic mass with a well-defined and slightly thick (up to 1 mm) capsular wall due to the presence of ovarian stroma. The majority of MCN-Ls are located near the hepatic hilum, especially in segment IV, likely due to dysembryogenesis (62).

Uniform curvilinear internal septa are present that measure less than 0.5 mm and at times may be better evaluated at US, especially with microbubble contrast agents, where the septa demonstrate enhancement (Fig E7). At CT and MRI, a thick enhancing capsule sometimes helps in differentiation from hepatic cysts, in addition to the presence of septa arising from the cyst wall without indentation and septal enhancement (63). At times, internal loculi resembling daughter cysts similar to those of hydatid disease can be identified (Fig 15).

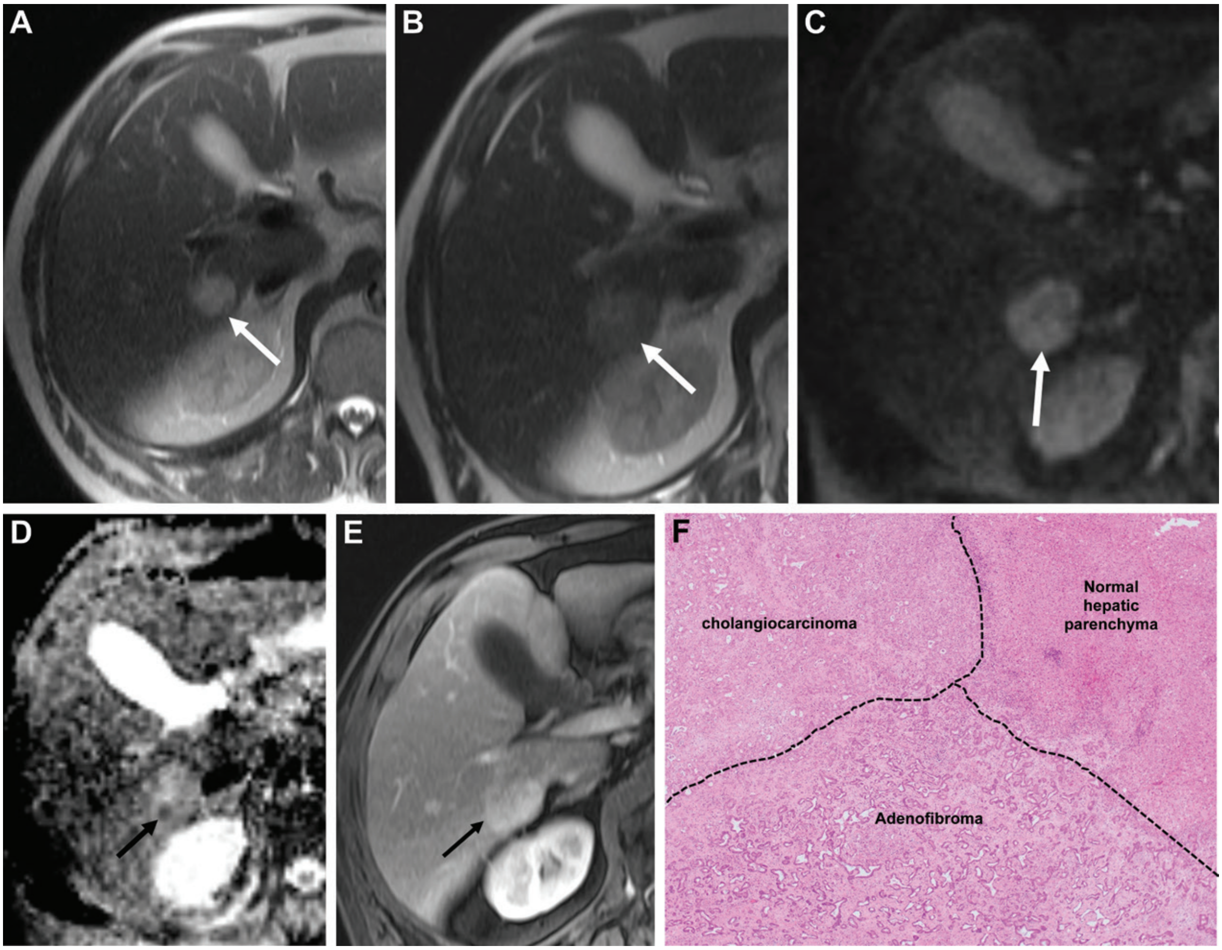

Figure 14. Adenofibroma with malignant transformation in an 80-year-old man initially presenting with vague right upper quadrant discomfort. **(A)** Axial HASTE image shows a T2-hyperintense mass (arrow) in the right hemiliver, which also demonstrated diffusion restriction and hyperenhancement (not shown). The mass was deemed indeterminate, warranting biopsy; histopathologic analysis demonstrated adenofibroma. Follow-up MRI was performed 1 year later. **(B)** Axial HASTE image shows increased size of the mass (arrow) with decreased T2 signal intensity. **(C, D)** Axial diffusion-weighted image **(C)** and ADC map **(D)** show asymmetric increased diffusion restriction (arrow) along the posterior lateral margin of the tumor. **(E)** Axial contrast-enhanced T1-weighted VIBE image shows arterial hyperenhancement of the tumor (arrow), with persistent delayed enhancement and no washout (not shown). **(F)** Photomicrograph after surgical resection shows malignant transformation into CCA. (H-E stain; original magnification, ×40.)

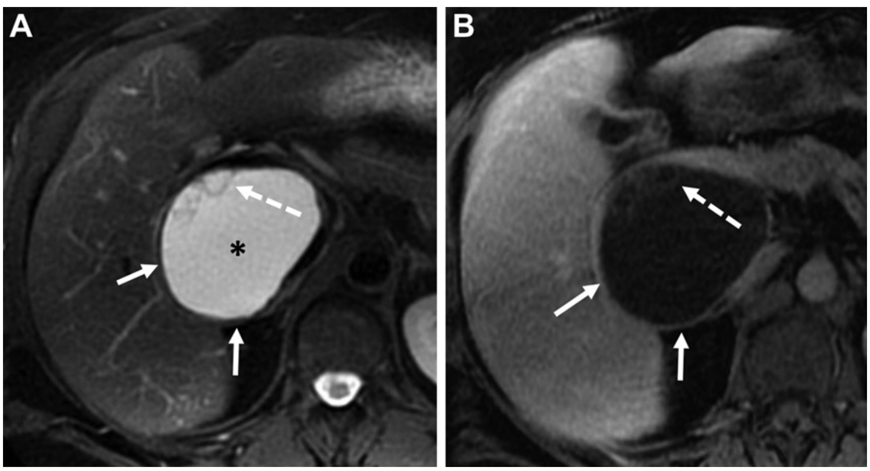

Figure 15. MCN-L in a 44-year-old woman with right abdominal pain. **(A)** Axial fat-saturated T2-weighed image shows a partially exophytic multiloculated cystic lesion (*) in the right hemiliver with a T2-hypointense thick wall (solid arrows) and irregular internal septa (dashed arrow) along the wall. **(B)** Axial T1-weighted image with intravenous contrast material in the portal venous phase shows enhancement of the wall (solid arrows) and septa (dashed arrow). Right hepatectomy was performed; histopathologic analysis demonstrated a mucinous cystic neoplasm with foci of epithelial atypia.

Fluid aspiration results can be variable, with increased CA 19-9 level seen with both MCN-L and simple hepatic cysts. The best discriminant is intracystic CA 72-4 level greater than 25 U/mL (25 kU/L) in MCN-L (64). The contents of MCN-L can be serous or mucinous fluid mixed

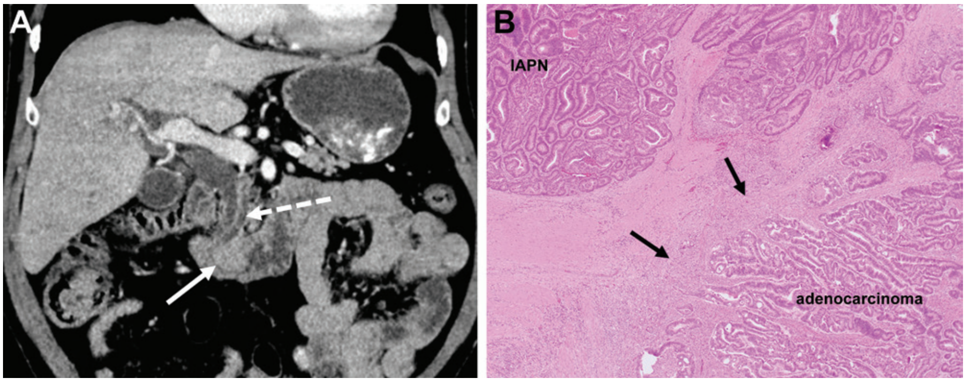

Figure 16. IAPN in a 61-year-old man with right upper quadrant abdominal pain and jaundice. **(A)** Coronal contrast-enhanced portal venous phase CT image shows a polypoid mass (solid arrow) at the ampulla with upstream biliary and pancreatic duct dilatation (dashed arrow)—the double duct sign. Initial histopathologic analysis from biopsy via upper endoscopy demonstrated IAPN. **(B)** Photomicrograph after the Whipple surgery shows areas of IAPN confined within the basement membrane of glands, in addition to foci of invasive adenocarcinoma extending into the lamina propria and causing a stromal reaction (arrows). (H-E stain; original magnification, ×40.)

with varying degrees of protein and hemorrhage, which can differ in attenuation and signal intensity at CT and MRI.

MCN-L typically does not communicate with the bile ducts, unlike IPNB, which may communicate (55). Papillary excrescences or polypoid solid mural nodularity is rare with benign MCN-L and should be concerning for malignant MCN-L, which accounts for 17%–19% of cystic liver tumors (65,66). Multiple thick septa along with coarse thick or capsular calcifications are more frequently seen with malignant MCN-L than with benign MCN-L.

Aspiration, unroofing, or fenestration of MCN-L is associated with a 45.6% recurrence rate. Complete surgical resection is the treatment of choice. For poor surgical candidates, periodic follow-up imaging can be performed (63).

Ampulla

Intra-ampullary Papillary Tubular Neoplasm.— *Intra-ampullary papillary tubular neoplasm* is the recent unified nomenclature proposed for mass-forming preinvasive ampullary lesions based on histopathologic features. Intra-ampullary papillary tubular neoplasm (IAPN) arises in the ampulla of Vater and has a prevalence of 0.04%–0.12%. The term *intra-ampullary papillary tubular neoplasm* encompasses previously known adenomas, and IAPN is considered a precursor lesion for ampullary adenocarcinoma.

In one large series, the majority of the IAPNs (~78%) harbored an invasive component (67). Approximately 28% of adenocarcinomas of the ampulla arise in association with IAPN and carry a better prognosis than typical invasive ampullary adenocarcinoma (67,68). Owing to the location and obstructive symptoms, this tumor declares itself at an early stage and may be surgically resectable at manifestation. However, prospective imaging diagnosis and differentiation for adenocarcinoma is often not possible.

At CT and MRI, a discrete polypoid soft-tissue mass is seen in the periampullary region, but it may be challenging to identify if smaller than 1 cm owing to duodenal folds. There may be subtle irregularity of the ampullary margin at CT and MRCP. Visualization can be improved when the duodenum is well distended with fluid. The double duct sign is present in 52% of cases, with both biliary and pancreatic duct dilatation prompting further evaluation with ERCP (Fig 16) (69). Surgery is the mainstay of treatment, with transduodenal ampullectomy or pancreatoduodenectomy.

Gallbladder

Intracholecystic Papillary Neoplasm.—First described in the 2010 World Health Organization (WHO) classification, intracholecystic papillary neoplasm (ICPN) shares clinicopathologic features with intraductal papillary mucinous neoplasm (IPMN) of the pancreas and IPNB. The term *intracholecystic papillary neoplasm* includes neoplastic polyp adenomas, as well as papillary neoplasms of the gallbladder larger than 1 cm, regardless of phenotype. These account for 0.5%–0.6% of gallbladder neoplasms and are considered premalignant, with 50%–75% of lesions demonstrating an invasive malignant component at histologic analysis (50).

The 5-year survival rate of patients with noninvasive ICPN is 78%–90%. Patients with ICPN and invasive carcinoma have significantly

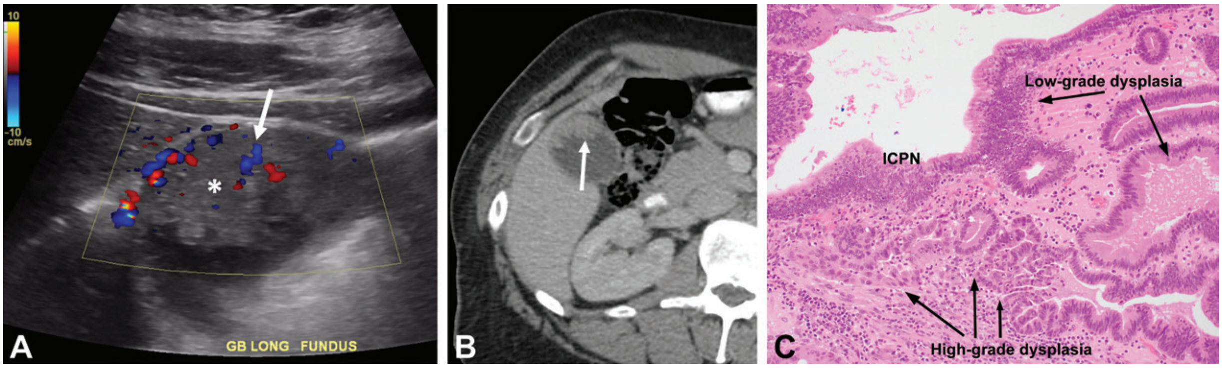

Figure 17. ICPN in a 56-year-old man with right upper quadrant pain. **(A)** Longitudinal Doppler image of the gallbladder shows an isoechoic mass (*) with an irregular papillary surface along the gallbladder fundus and confined to the gallbladder lumen. Internal vascularity is present, as evidenced by the detectable Doppler flow (arrow). **(B)** Axial contrast-enhanced portal venous phase CT image shows internal enhancement (arrow) within the mass. Cholecystectomy was performed. **(C)** Photomicrograph shows an ICPN with extensive high-grade dysplasia. (H-E stain; original magnification, ×40).

better clinical outcome than those with typical invasive gallbladder cancer, likely owing to an endophytic growth pattern into the gallbladder lumen that increases the chance of diagnosis at an early stage and curative resection (70). Microscopically, ICPN shows a papillary or tubulopapillary growth pattern, often with high-grade dysplasia (71).

At US, ICPN appears as a sessile polypoid soft-tissue mass along the gallbladder wall with varying degrees of discernible internal vascularity. The surface of the mass is often irregular owing to papillary projections (Fig 17). Approximately 20% of cases are associated with gallstones. ICPN can appear as an irregular mass, which makes it difficult to prospectively differentiate it from malignancy, leading to cholecystectomy and diagnosis.

At CT and MRI, single or multiple filling defects along the gallbladder wall can be seen, with strong enhancement after contrast material administration and diffusion restriction. A hypointense stalk may be identified on T2-weighted images (72). Gallbladder wall discontinuity and a mass extending into the adjacent liver parenchyma or omentum are indicative of invasive malignancy.

Conclusion

Certain congenital, inflammatory, and neoplastic conditions predispose individuals to biliary malignancy. Radiologist should be aware of the various imaging features of these conditions to correctly diagnose and indicate possible malignant transformation to facilitate early diagnosis and potential curative management.

***Disclosures of conflicts of interest.*—A.B.S.** Consulting fees and honoraria for lectures from Bayer. **M.R.B.** Grants from Carmot Therapeutics, Corcept Therapeutics, Madrigal Pharmaceuticals, Metacrine, NGM Biopharmaceuticals, and Pinnacle Clinical Research; consulting fees from Corcept Therapeutics, ICON, and Medpace. **K.F.** Grants or research support from General Electric, Siemens, Bayer, Median, and Pfizer; consulting fees from IBM and Epigenomics.

References

1. Mizuguchi Y, Specht S, Isse K, Lunz JG, Demetris AJ. Biliary Epithelial Cells. In: Monga SPS, ed. Molecular Pathology of Liver Diseases. Boston, Mass: Springer, 2011; 27–51.
2. Alpini G, McGill JM, Larusso NF. The pathobiology of biliary epithelia. Hepatology 2002;35(5):1256–1268.
3. de Jong IEM, van Leeuwen OB, Lisman T, Gouw ASH, Porte RJ. Repopulating the biliary tree from the peribiliary glands. Biochim Biophys Acta Mol Basis Dis 2018;1864(4 Pt B):1524–1531.
4. Cardinale V, Carpino G, Reid L, Gaudio E, Alvaro D. Multiple cells of origin in cholangiocarcinoma underlie biological, epidemiological and clinical heterogeneity. World J Gastrointest Oncol 2012;4(5):94–102.
5. Labib PL, Goodchild G, Pereira SP. Molecular Pathogenesis of Cholangiocarcinoma. BMC Cancer 2019;19(1):185.
6. Banales JM, Marin JJG, Lamarca A, et al. Cholangiocarcinoma 2020: the next horizon in mechanisms and management. Nat Rev Gastroenterol Hepatol 2020;17(9):557–588.
7. Desmet VJ. Congenital diseases of intrahepatic bile ducts: variations on the theme "ductal plate malformation." Hepatology 1992;16(4):1069–1083.
8. Bloustein PA. Association of carcinoma with congenital cystic conditions of the liver and bile ducts. Am J Gastroenterol 1977;67(1):40–46.
9. Singham J, Yoshida EM, Scudamore CH. Choledochal cysts: part 1 of 3—classification and pathogenesis. Can J Surg 2009;52(5):434–440.
10. Fieber SS, Nance FC. Choledochal cyst and neoplasm: a comprehensive review of 106 cases and presentation of two original cases. Am Surg 1997;63(11):982–987.
11. Todani T, Tabuchi K, Watanabe Y, Kobayashi T. Carcinoma arising in the wall of congenital bile duct cysts. Cancer 1979;44(3):1134–1141.
12. Ten Hove A, de Meijer VE, Hulscher JBF, de Kleine RHJ. Meta-analysis of risk of developing malignancy in congenital choledochal malformation. Br J Surg 2018;105(5):482–490.
13. Liu QY, Lai DM, Gao M, et al. MRI manifestations of adult choledochal cysts associated with biliary malignancy: a report of ten cases. Abdom Imaging 2013;38(5):1061–1070.
14. Yoshida H, Itai Y, Minami M, Kokubo T, Ohtomo K, Kuroda A. Biliary malignancies occurring in choledochal cysts. Radiology 1989;173(2):389–392.
15. Lewin M, Desterke C, Guettier C, et al. Diffuse Versus Localized Caroli Disease: A Comparative MRCP Study. AJR Am J Roentgenol 2021;216(6):1530–1538.
16. Levy AD, Rohrmann CA Jr, Murakata LA, Lonergan GJ. Caroli's disease: radiologic spectrum with pathologic correlation. AJR Am J Roentgenol 2002;179(4):1053–1057.

17. Baron RL, Campbell WL, Dodd GD 3rd. Peribiliary cysts associated with severe liver disease: imaging-pathologic correlation. AJR Am J Roentgenol 1994;162(3):631–636.
18. Chung YE, Kim MJ, Park YN, et al. Varying appearances of cholangiocarcinoma: radiologic-pathologic correlation. RadioGraphics 2009;29(3):683–700.
19. Kamisawa T, Kuruma S, Chiba K, Tabata T, Koizumi S, Kikuyama M. Biliary carcinogenesis in pancreaticobiliary maljunction. J Gastroenterol 2017;52(2):158–163.
20. Hamada Y, Ando H, Kamisawa T, et al. Diagnostic criteria for congenital biliary dilatation 2015. J Hepatobiliary Pancreat Sci 2016;23(6):342–346.
21. Ono A, Arizono S, Isoda H, Togashi K. Imaging of Pancreaticobiliary Maljunction. RadioGraphics 2020;40(2):378–392.
22. Kamisawa T, Honda G. Pancreaticobiliary Maljunction: Markedly High Risk for Biliary Cancer. Digestion 2019;99(2):123–125.
23. Wiesner RH, LaRusso NF. Clinicopathologic features of the syndrome of primary sclerosing cholangitis. Gastroenterology 1980;79(2):200–206.
24. Vitellas KM, Keogan MT, Freed KS, et al. Radiologic manifestations of sclerosing cholangitis with emphasis on MR cholangiopancreatography. RadioGraphics 2000;20(4):959–975; quiz 1108–1109, 1112.
25. Claessen MM, Vleggaar FP, Tytgat KM, Siersema PD, van Buuren HR. High lifetime risk of cancer in primary sclerosing cholangitis. J Hepatol 2009;50(1):158–164.
26. Venkatesh SK, Welle CL, Miller FH, et al. Reporting standards for primary sclerosing cholangitis using MRI and MR cholangiopancreatography: guidelines from MR Working Group of the International Primary Sclerosing Cholangitis Study Group. Eur Radiol 2021. 10.1007/s00330-021-08147-7. Published online August 6, 2021. [Published correction appears in Eur Radiol 2021 Oct 4. 10.1007/s00330-021-08333-7.]
27. Fulcher AS, Turner MA. HASTE MR cholangiography in the evaluation of hilar cholangiocarcinoma. AJR Am J Roentgenol 1997;169(6):1501–1505.
28. Kwan KEL, Shelat VG, Tan CH. Recurrent pyogenic cholangitis: a review of imaging findings and clinical management. Abdom Radiol (NY) 2017;42(1):46–56.
29. Huang MH, Chen CH, Yen CM, et al. Relation of hepatolithiasis to helminthic infestation. J Gastroenterol Hepatol 2005;20(1):141–146.
30. Catalano OA, Sahani DV, Forcione DG, et al. Biliary infections: spectrum of imaging findings and management. RadioGraphics 2009;29(7):2059–2080.
31. Heffernan EJ, Geoghegan T, Munk PL, Ho SG, Harris AC. Recurrent pyogenic cholangitis: from imaging to intervention. AJR Am J Roentgenol 2009;192(1):W28–W35.
32. Kim HJ, Kim JS, Joo MK, et al. Hepatolithiasis and intrahepatic cholangiocarcinoma: a review. World J Gastroenterol 2015;21(48):13418–13431.
33. Yu TH, Yuan RH, Chen YL, Yang WC, Hsu HC, Jeng YM. Viral hepatitis is associated with intrahepatic cholangiocarcinoma with cholangiolar differentiation and N-cadherin expression. Mod Pathol 2011;24(6):810–819.
34. Lee CH, Chang CJ, Lin YJ, Yeh CN, Chen MF, Hsieh SY. Viral hepatitis-associated intrahepatic cholangiocarcinoma shares common disease processes with hepatocellular carcinoma. Br J Cancer 2009;100(11):1765–1770.
35. Choi SH, Lee SS, Park SH, et al. LI-RADS Classification and Prognosis of Primary Liver Cancers at Gadoxetic Acid–enhanced MRI. Radiology 2019;290(2):388–397.
36. Fowler KJ, Potretzke TA, Hope TA, Costa EA, Wilson SR. LI-RADS M (LR-M): definite or probable malignancy, not specific for hepatocellular carcinoma. Abdom Radiol (NY) 2018;43(1):149–157.
37. Chernyak V, Fowler KJ, Kamaya A, et al. Liver Imaging Reporting and Data System (LI-RADS) Version 2018: Imaging of Hepatocellular Carcinoma in At-Risk Patients. Radiology 2018;289(3):816–830.
38. Lu CD, Wang K, Zhang CZ, et al. Outcomes of intrahepatic cholangiocarcinoma with portal vein tumor thrombus following hepatic resection. J Gastroenterol Hepatol 2016;31(7):1330–1335.
39. Fiorotto R, Strazzabosco M. Cystic Fibrosis-Related Liver Diseases: New Paradigm for Treatment Based on Pathophysiology. Clin Liver Dis (Hoboken) 2016;8(5):113–116.
40. King LJ, Scurr ED, Murugan N, Williams SGJ, Westaby D, Healy JC. Hepatobiliary and pancreatic manifestations of cystic fibrosis: MR imaging appearances. RadioGraphics 2000;20(3):767–777.
41. Gillespie CD, O'Reilly MK, Allen GN, McDermott S, Chan VO, Ridge CA. Imaging the Abdominal Manifestations of Cystic Fibrosis. Int J Hepatol 2017;2017:5128760.
42. Lavelle LP, McEvoy SH, Ni Mhurchu E, et al. Cystic Fibrosis below the Diaphragm: Abdominal Findings in Adult Patients. RadioGraphics 2015;35(3):680–695.
43. Maisonneuve P, Marshall BC, Knapp EA, Lowenfels AB. Cancer risk in cystic fibrosis: a 20-year nationwide study from the United States. J Natl Cancer Inst 2013;105(2):122–129.
44. Khan ZS, Livingston EH, Huerta S. Reassessing the need for prophylactic surgery in patients with porcelain gallbladder: case series and systematic review of the literature. Arch Surg 2011;146(10):1143–1147.
45. DesJardins H, Duy L, Scheirey C, Schnelldorfer T. Porcelain Gallbladder: Is Observation a Safe Option in Select Populations? J Am Coll Surg 2018;226(6):1064–1069.
46. Stephen AE, Berger DL. Carcinoma in the porcelain gallbladder: a relationship revisited. Surgery 2001;129(6):699–703.
47. Lim JH, Jang KT, Kim JH. Precursors of Cholangiocarcinoma. In: Neoplasms of the Biliary Tract: Radiologic and Pathologic Correlations. Singapore: Springer, 2021; 49–58.
48. Zen Y, Adsay NV, Bardadin K, et al. Biliary intraepithelial neoplasia: an international interobserver agreement study and proposal for diagnostic criteria. Mod Pathol 2007;20(6):701–709.
49. Sibulesky L, Nguyen J, Patel T. Preneoplastic conditions underlying bile duct cancer. Langenbecks Arch Surg 2012;397(6):861–867.
50. Klöppel G, Adsay V, Konukiewitz B, Kleeff J, Schlitter AM, Esposito I. Precancerous lesions of the biliary tree. Best Pract Res Clin Gastroenterol 2013;27(2):285–297.
51. Park HJ, Kim SY, Kim HJ, et al. Intraductal Papillary Neoplasm of the Bile Duct: Clinical, Imaging, and Pathologic Features. AJR Am J Roentgenol 2018;211(1):67–75.
52. Lim JH, Jang KT, Choi D. Biliary intraductal papillary-mucinous neoplasm manifesting only as dilatation of the hepatic lobar or segmental bile ducts: imaging features in six patients. AJR Am J Roentgenol 2008;191(3):778–782.
53. Yoon HJ, Kim YK, Jang KT, et al. Intraductal papillary neoplasm of the bile ducts: description of MRI and added value of diffusion-weighted MRI. Abdom Imaging 2013;38(5):1082–1090.
54. Lim JH, Yoon KH, Kim SH, et al. Intraductal papillary mucinous tumor of the bile ducts. RadioGraphics 2004;24(1):53–66; discussion 66–67.
55. Zen Y, Fujii T, Itatsu K, et al. Biliary cystic tumors with bile duct communication: a cystic variant of intraductal papillary neoplasm of the bile duct. Mod Pathol 2006;19(9):1243–1254.
56. Tsui WM, Loo KT, Chow LT, Tse CC. Biliary adenofibroma: a heretofore unrecognized benign biliary tumor of the liver. Am J Surg Pathol 1993;17(2):186–192.
57. Sturm AK, Welsch T, Meissner C, Aust DE, Baretton G. A case of biliary adenofibroma of the liver with malignant transformation: a morphomolecular case report and review of the literature. Surg Case Rep 2019;5(1):104.
58. Lee S, Kim KW, Jeong WK, Yu E, Jang KT. Magnetic Resonance Imaging Findings of Biliary Adenofibroma. Korean J Gastroenterol 2019;74(6):356–361.
59. Akin O, Coskun M. Biliary adenofibroma with malignant transformation and pulmonary metastases: CT findings. AJR Am J Roentgenol 2002;179(1):280–281.
60. Lim JH, Jang KT, Kim JH. Cystic Neoplasms. In: Neoplasms of the Biliary Tract: Radiologic and Pathologic Correlations. Singapore: Springer, 2021; 27–40.
61. Hamaoka M, Kozaka K, Matsui O, et al. Early detection of intrahepatic cholangiocarcinoma. Jpn J Radiol 2019;37(10):669–684.
62. Mortelé KJ, Ros PR. Cystic focal liver lesions in the adult: differential CT and MR imaging features. RadioGraphics 2001;21(4):895–910.

63. Boyum JH, Sheedy SP, Graham RP, et al. Hepatic Mucinous Cystic Neoplasm Versus Simple Biliary Cyst: Assessment of Distinguishing Imaging Features Using CT and MRI. AJR Am J Roentgenol 2021;216(2):403–411.
64. Fuks D, Voitot H, Paradis V, Belghiti J, Vilgrain V, Farges O. Intracystic concentrations of tumour markers for the diagnosis of cystic liver lesions. Br J Surg 2014;101(4):408–416.
65. Wheeler DA, Edmondson HA. Cystadenoma with mesenchymal stroma (CMS) in the liver and bile ducts: a clinicopathologic study of 17 cases, 4 with malignant change. Cancer 1985;56(6):1434–1445.
66. Akwari OE, Tucker A, Seigler HF, Itani KM. Hepatobiliary cystadenoma with mesenchymal stroma. Ann Surg 1990;211(1):18–27.
67. Ohike N, Kim GE, Tajiri T, et al. Intra-ampullary papillary-tubular neoplasm (IAPN): characterization of tumoral intraepithelial neoplasia occurring within the ampulla: a clinicopathologic analysis of 82 cases. Am J Surg Pathol 2010;34(12):1731–1748.
68. Xue Y, Vanoli A, Balci S, et al. Non-ampullary-duodenal carcinomas: clinicopathologic analysis of 47 cases and comparison with ampullary and pancreatic adenocarcinomas. Mod Pathol 2017;30(2):255–266.
69. Nikolaidis P, Hammond NA, Day K, et al. Imaging features of benign and malignant ampullary and periampullary lesions. RadioGraphics 2014;34(3):624–641.
70. Adsay V, Jang KT, Roa JC, et al. Intracholecystic papillary-tubular neoplasms (ICPN) of the gallbladder (neoplastic polyps, adenomas, and papillary neoplasms that are ≥1.0 cm): clinicopathologic and immunohistochemical analysis of 123 cases. Am J Surg Pathol 2012;36(9):1279–1301.
71. Mizobuchi N, Munechika J, Takeyama N, et al. Three cases of intracystic papillary neoplasm of gallbladder. Abdom Radiol (NY) 2018;43(7):1535–1539.
72. Chatterjee A, Lopes Vendrami C, Nikolaidis P, et al. Uncommon Intraluminal Tumors of the Gallbladder and Biliary Tract: Spectrum of Imaging Appearances. RadioGraphics 2019;39(2):388–412.

*This journal-based SA-CME activity has been approved for **AMA PRA Category 1 Credit™**. See rsna.org/learning-center-rg.*